AI AgriYield Predictor

Milestone 1 Documentation

Requirements & Dataset Preparation

Title

AI-Based Crop Yield Prediction Using Soil and Weather Parameters.

Project Objective

The objective of this project is to develop a machine learning model to predict crop yield (kg per acre) using soil nutrients, weather conditions, fertilizer usage, and weather-related features.

Since the target variable (Yield_kg_per_acre) is a continuous numerical value, this problem is treated as a regression problem.

Data Source

The dataset was provided by the project mentor as part of the Infosys Springboard Virtual Internship.

Dataset details:

- Total records: 1500
- Total columns: 12
- Categorical features: State, Crop, Soil_Type, Fertilizer
- Numerical features: N, P, K, Rainfall_mm, Temperature_C, Soil_pH, Year
- Target variable: Yield_kg_per_acre

Process Followed

Step 1: Environment Setup

- Created a virtual environment.
- Installed required libraries (pandas, numpy, seaborn, matplotlib, scikit-learn).

- Used Jupyter Notebook for implementation.

Step 2: Data Exploration

The dataset was loaded using:

```
df = pd.read_csv("dataset.csv")
df.head()
```

	State	Crop	Soil_Type	Fertilizer	N	P	K	Rainfall_mm	Temperature_C	Yield_kg_per_acre	Soil_pH	Year
0	Karnataka	Soybean	Loamy	DAP	96	41	51	123	31.06	1899	6.82	2003
1	Odisha	Cotton	Red Soil	Urea	29	36	112	247	33.97	1002	6.41	2002
2	Punjab	Groundnut	Red Soil	Compost	37	38	177	142	24.21	1465	7.06	2015
3	Gujarat	Wheat	Red Soil	Compost	58	77	129	227	30.85	2273	5.93	2022
4	Andhra Pradesh	Cotton	Clay	Organic	108	61	63	263	37.81	1497	6.24	2017

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   State              1500 non-null   str
 1   Crop               1500 non-null   str
 2   Soil_Type          1500 non-null   str
 3   Fertilizer         1500 non-null   str
 4   N                  1500 non-null   int64
 5   P                  1500 non-null   int64
 6   K                  1500 non-null   int64
 7   Rainfall_mm        1500 non-null   int64
 8   Temperature_C      1500 non-null   float64
 9   Yield_kg_per_acre  1500 non-null   int64
 10  Soil_pH            1500 non-null   float64
 11  Year               1500 non-null   int64
dtypes: float64(2), int64(6), str(4)
memory usage: 140.8 KB
```

```
df.describe()
```

	N	P	K	Rainfall_mm	Temperature_C	Yield_kg_per_acre	Soil_pH	Year
count	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000	1500.000000
mean	73.438000	61.282667	102.112667	176.806000	28.148547	7881.892667	6.766347	2012.090000
std	37.756932	33.257422	54.362863	73.776182	5.783160	20073.333665	0.724898	7.393218
min	10.000000	5.000000	10.000000	50.000000	18.020000	502.000000	5.500000	2000.000000
25%	41.000000	33.000000	56.000000	113.000000	23.132500	1202.750000	6.120000	2005.000000
50%	70.000000	61.000000	100.000000	176.000000	28.490000	1673.500000	6.795000	2012.000000
75%	106.000000	91.000000	149.000000	242.000000	33.180000	2571.750000	7.400000	2019.000000
max	139.000000	119.000000	199.000000	299.000000	37.960000	89946.000000	8.000000	2024.000000

Observations:

- Dataset contains 1500 rows and 12 columns.
- No missing values were found.
- Data types were appropriate for processing.
- Yield variable showed slight skewness due to some high maximum values.

Step 3: Data Cleaning

Missing values were checked using:

```
df.isnull().sum()
```

Duplicate rows were checked using:

```
duplicates = df.duplicated().sum()
```

Results:

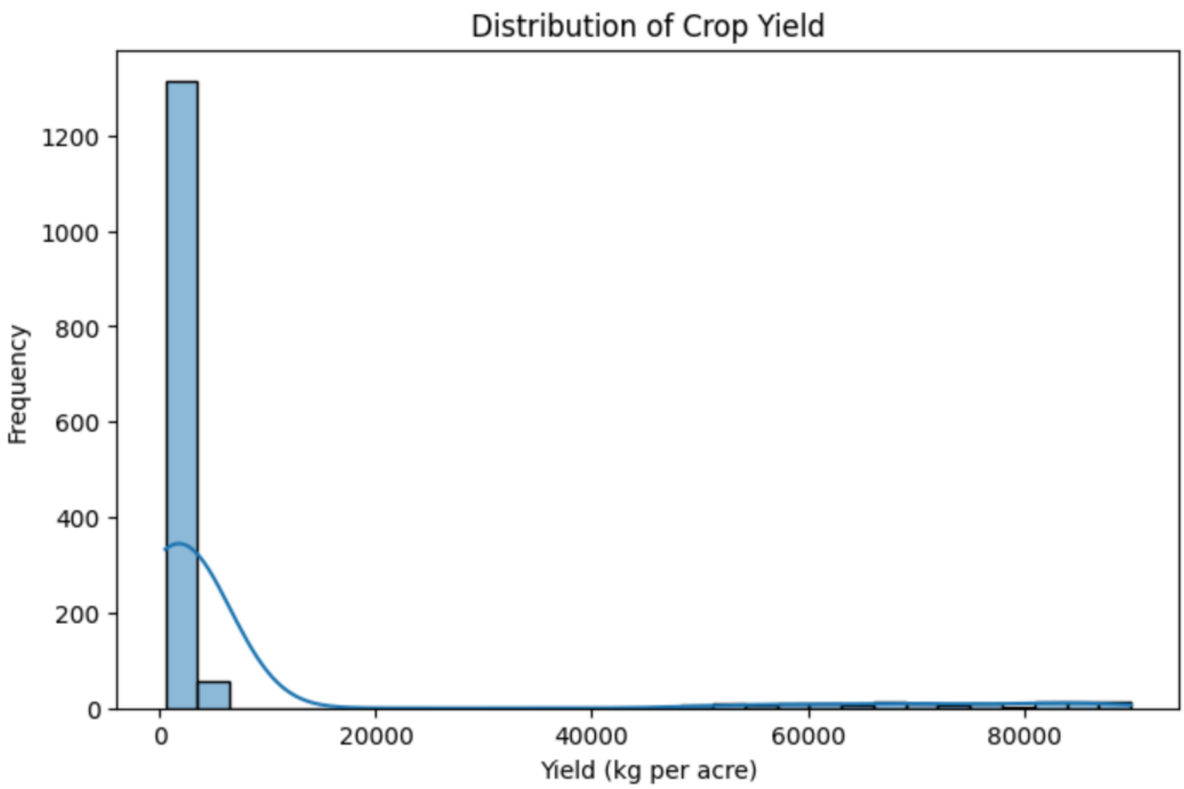
- No missing values
- No duplicate records

Thus, no additional cleaning was required.

Data Analysis (Graphs)

Graph 1: Distribution of Crop Yield

```
plt.figure(figsize=(8,5))

sns.histplot(df['Yield_kg_per_acre'], bins=30, kde=True)

plt.title("Distribution of Crop Yield")

plt.xlabel("Yield (kg per acre)")

plt.ylabel("Frequency")

plt.show()
```

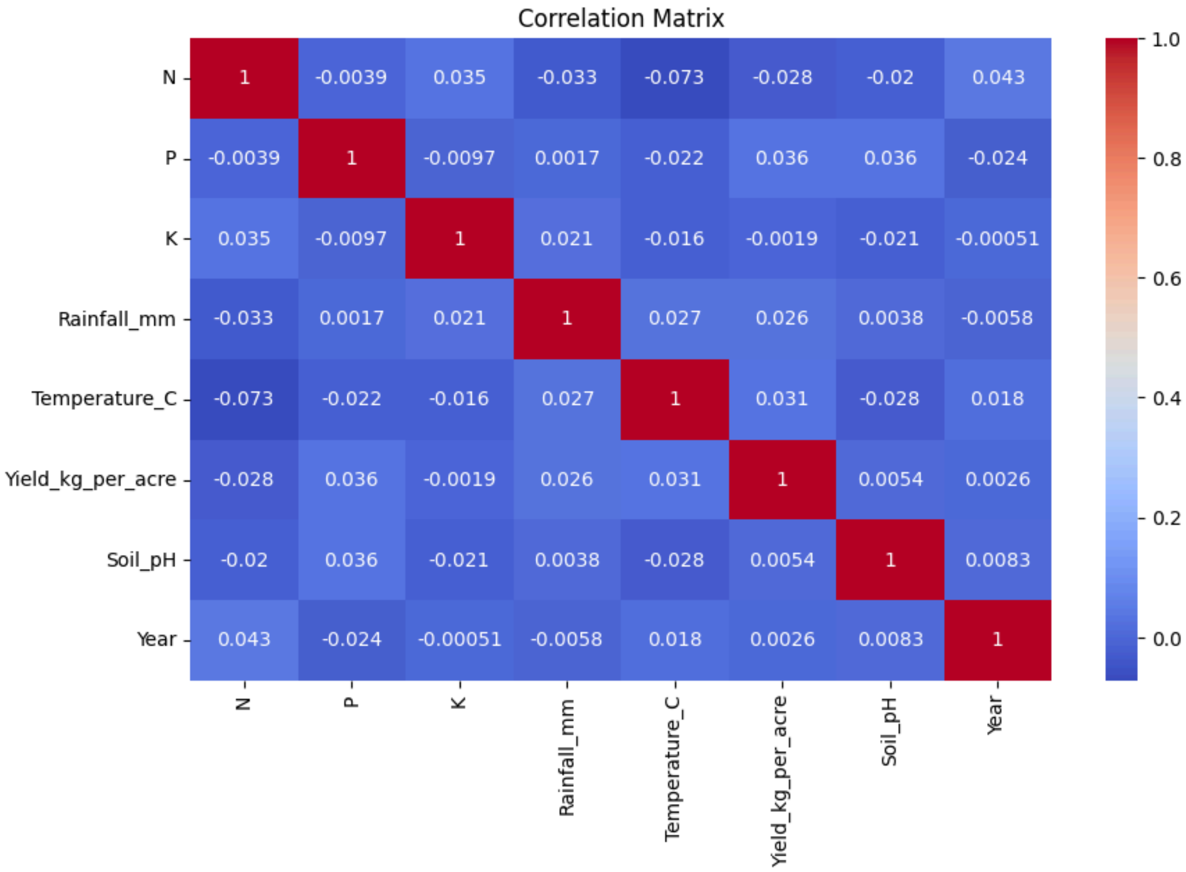


Observation:

- The yield distribution is slightly right-skewed.
- Some crops have significantly higher yield values compared to the majority.
- Most yield values fall within a moderate range.

Graph 2: Correlation Heatmap

```python
plt.figure(figsize=(10,6))

sns.heatmap(df.corr(numeric_only=True), annot=True,
cmap='coolwarm')

plt.title("Correlation Matrix")

plt.show()
```



Observation:

- The heatmap shows relationships between numerical features.
- Some soil nutrients and rainfall show correlation with yield.
- Helps in understanding which features may influence crop productivity.

Feature Engineering

One-Hot Encoding

Since machine learning models cannot process categorical text data directly, categorical features were converted using one-hot encoding:

```
X = df.drop("Yield_kg_per_acre", axis=1)

y = df["Yield_kg_per_acre"]

X_encoded = pd.get_dummies(X, drop_first=True)
```

The dataset was split into features (X) and target variable (y) before applying encoding.

Result:

- Number of features increased from 12 to 39.
- All categorical variables were converted into numerical format.

Feature Scaling

Numerical features had different ranges (e.g., rainfall in hundreds, pH in single digits).To ensure uniform scaling of numerical features, StandardScaler was applied only to the following columns:

```
scaler = StandardScaler()

numerical_cols = ['N', 'P', 'K', 'Rainfall_mm',
'Temperature_C', 'Soil_pH', 'Year']

X_encoded[numerical_cols] =
scaler.fit_transform(X_encoded[numerical_cols])

print("Final dataset shape after preprocessing:",
X_encoded.shape)
```

After scaling:

- Numerical features were standardized (mean ~ 0, std ~ 1)
- Dataset shape became (1500, 39)
- No rows were lost during preprocessing

Challenges Faced

1. Initial virtual environment activation issues.
2. Execution delay due to kernel configuration.
3. Understanding the correct order of preprocessing steps.

These were resolved through proper environment setup and debugging.

Outcome of Milestone 1

- Dataset was explored and understood.
- No missing or duplicate data.
- Categorical features were encoded.
- Numerical features were standardized.
- 2 important visualizations were generated.
- Final dataset prepared for regression model training.

Conclusion

Milestone 1 successfully completed dataset preparation and preprocessing steps required for machine learning. The data is now clean, structured, encoded, and scaled, making it ready for building regression models in the next milestone.

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